03 Continuity

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1 The Concept of Continuity (Sequential Definition)

1.1 The Logic & Spec View (QA approach)

- **Goal:** To ensure the system preserves "stability" or "convergence." If inputs settle down, outputs must also settle down to the predicted value.
- **The Test (Input):** A sequence of inputs (x_n) converging to a specific target $\hat{x}$.
- **The Expectation:** The corresponding sequence of outputs $f(x_n)$ must converge to $f(\hat{x})$.
- **Failure Condition:** If the input approaches $\hat{x}$ but the output jumps to a value different from $f(\hat{x})$ (or oscillates), the function is "discontinuous" or "broken" at that point.
- **Algorithmic Implication:** You can compute the limit of the function by just plugging in the limit of the sequence. $f(\lim x_n) = \lim f(x_n)$. The operation "commutes" with the limit.

1.2 The Intuitive Explanation

- **No Teleporting:** Imagine walking along the number line towards a specific point. On the graph, your height (function value) should approach the height of the destination. You shouldn't suddenly "warp" to a different height the moment you arrive.
- **Predictability:** If you know where you are going, you should know where you will end up. Continuity guarantees no surprises at the destination.

1.3 Formal Definition

Let (Ω, d_Ω) and $(\tilde{\Omega}, d_{\tilde{\Omega}})$ be metric spaces. A function $f : \Omega \to \tilde{\Omega}$ is **continuous at** $\hat{x} \in \Omega$ if for every sequence $(x_n)_{n\in\mathbb{N}} \subseteq \Omega$ with $x_n \to \hat{x}$:

$$\lim_{n\to\infty} f(x_n) = f(\hat{x})$$

We write $\lim_{x\to\hat{x}} f(x) = f(\hat{x})$.

2 Examples of Continuity and Discontinuity

2.1 Polynomials vs. Piecewise Functions

2.1.1 The Logic & Spec View

- **Polynomials ($2x^2 + 3$):** Built from basic arithmetic operations (multiplication, addition) which are themselves continuous. They pass the stability test everywhere.
- **Piecewise (Step Functions):** These contain "conditional branching" (if $x \geq 0$ do A, else do B).
- **The Bug:** At the branching point ($x = 0$), approaching from the left (negative) executes logic B, approaching from the right (positive) executes logic A. If $A \neq B$, the system crashes (discontinuity).

2.1.2 The Intuitive Explanation

- **Polynomial:** A roller coaster track. Smooth, no breaks.
- **Piecewise Discontinuity:** A drawbridge that is stuck open. The road exists on both sides, but they don't meet in the middle. Approaching 0 from -0.0001 puts you at height 2. Arriving at 0 puts you at height 1. That's a vertical jump (or gap).

2.1.3 Formal Verification

- For $f(x) = 2x^2 + 3$: If $x_n \to 2$, then $f(x_n) \to 2(2^2) + 3 = 11 = f(2)$. (Continuous).
- For $f(x) = \begin{cases} 2x + 1 & x \geq 0 \\ 4x^2 + 2 & x < 0 \end{cases}$: Let $x_n = -1/n \to 0$. Then $f(x_n) = 4(-1/n)^2 + 2 \to 2$. But $f(0) = 2(0) + 1 = 1$. Since $2 \neq 1$, f is discontinuous at 0.

3 Composition of Functions

3.1 The Logic & Spec View

- **Concept:** Chaining systems. System A pipes data into System B.

- **Logic:** If System A is stable (continuous) and System B is stable (continuous), the combined pipeline $B(A(x))$ is stable.
- **Verification:**
  1. Input $x_n \to x$.
  2. Since f is continuous, intermediate output $y_n = f(x_n) \to f(x) = y$.
  3. Since g is continuous, final output $g(y_n) \to g(y)$.
  4. Result: $g(f(x_n)) \to g(f(x))$.

3.2 The Intuitive Explanation

- If a map is continuous (no tearing), and you draw a path on it continuously, the projection of that path onto a globe is also continuous. "Smooth plus smooth equals smooth."

3.3 Formal Lemma

If $f : \Omega \to \tilde{\Omega}$ and $g : \tilde{\Omega} \to \Omega'$ are continuous, then $g \circ f : \Omega \to \Omega'$ is continuous.

4 Intermediate Value Theorem (IVT)

4.1 The Logic & Spec View

- **Goal:** Existence check. Can we guarantee a solution (a zero) exists?
- **Pre-conditions (Inputs):**
  1. A continuous process f on a closed interval $[a, b]$.
  2. A sign change: Start negative ($f(a) < 0$), end positive ($f(b) > 0$).
- **Guarantee (Output):** There exists at least one input ξ where the output is exactly zero.
- **Algorithm (Bisection):** This is the theoretical basis for Binary Search. You can chop the interval in half repeatedly to locate the zero.

4.2 The Intuitive Explanation

- **The River Crossing:** If you are on the south bank (negative) and want to get to the north bank (positive), and you cannot fly or teleport (function is continuous), you **must** get wet (cross the water at 0) at some point.

4.3 Formal Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous.

$$f(a) < 0 \wedge f(b) > 0 \implies \exists \xi \in (a, b) \text{ such that } f(\xi) = 0$$

Corollary: If $f(a) < y < f(b)$, there exists $x \in (a, b)$ such that $f(x) = y$. (Image of an interval is an interval).

5 Compactness (Min-Max Theorem)

5.1 The Logic & Spec View

- **Goal:** Boundary analysis. Does the system have a definitive "worst case" (minimum) and "best case" (maximum)?
- **Pre-conditions:**
  1. Continuous function.
  2. "Compact" domain: Closed and Bounded interval $[a, b]$.
- **Edge Cases:** If the interval is open $(0, 1)$ or infinite $[0, \infty)$, the max might be approached but never reached (like $1/x$ approaching ∞ or x^2 on $(0, 1)$ approaching 1).
- **Guarantee:** You will actually hit the maximum and minimum values at specific points p, q inside the domain.

5.2 The Intuitive Explanation

- **Physical Limits:** If you drive a car for a fixed time on a continuous road, there was a specific moment where you were going fastest, and a specific moment where you were slowest. You can't just "get faster and faster forever" (unbounded) or "approach 60mph but never hit it" (open set).

5.3 Formal Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous. There exist $p, q \in [a, b]$ such that:

$$f(p) = \sup_{x \in [a,b]} f(x) \quad \text{and} \quad f(q) = \inf_{x \in [a,b]} f(x)$$

6 Epsilon-Delta Definition (Alternative View)

6.1 The Logic & Spec View

- **Goal:** To define continuity without relying on sequences (time), but using "proximity" (space).
- **The Challenge (ϵ):** User specifies a tolerance for error in the output. "Output must be within $\pm\epsilon$ of the target."
- **The Response (δ):** Engineer specifies the required precision of the input. "Input must be within $\pm\delta$ of the source."
- **The Loop:** For **any** error tolerance ϵ, **there exists** a precision setting δ.

6.2 The Intuitive Explanation

- **Target Practice:** To hit the bullseye (output) within a certain ring size (ϵ), how steady (close to center) does your hand (input) need to be (δ)? If you can always steady your hand enough to hit any size ring, you are a "continuous" shooter.

6.3 Formal Lemma

f is continuous at x if and only if:

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } d_\Omega(x, y) < \delta \implies d_{\tilde{\Omega}}(f(x), f(y)) < \epsilon$$

7 Inverse Functions

7.1 The Logic & Spec View

- **Goal:** Reversibility. Can we run the system backwards?

- **Condition:** The function must be strictly strictly monotone (always increasing or always decreasing). This ensures a 1-to-1 mapping (no collisions).
- **Result:** If f is continuous and strictly monotone on an interval, the inverse f^{-1} is also continuous.

7.2 The Intuitive Explanation

- **Climbing a Ladder:** If you climb up a ladder without ever stepping down (strictly increasing), every rung corresponds to a unique height. You can look at a height and know exactly which rung you are on. If the ladder is unbroken (continuous), the path back down is also unbroken.

7.3 Formal Proposition

If $f : [a, b] \to [A, B]$ is continuous and strictly monotonically increasing, then f is bijective and $f^{-1} : [A, B] \to [a, b]$ is continuous.

8 The Logarithm and Powers

8.1 The Logic & Spec View

- **Exponential (exp):** We established it's continuous and strictly increasing.
- **Logarithm (log):** Defined as the inverse of exp. By the Inverse Function theorem above, log is automatically continuous.
- **Power (a^x):** Defined via composition: $a^x := \exp(x \log a)$. Since exp, multiplication, and log are continuous, their composition a^x is continuous.

8.2 The Intuitive Explanation

- **Tool Building:** We built the "Hammer" (Exponential). We used the Hammer to build the "Nail Puller" (Logarithm). We used both to build the "Power Drill" (a^x). Since the parts are solid, the tools are solid.

8.3 Formal Definition (Power Function)

For $a > 0$ and $x \in \mathbb{R}$:

$$a^x := \exp(x \log a)$$

Properties derived from continuity and functional equations:

- $a^{x+y} = a^x a^y$
- $\log(a^x) = x \log a$
- $(a^x)^y = a^{xy}$